



THE COPPERBELT UNIVERSITY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE
LITERATURE REVIEW

**PROJECT TITLE: THE KEY PERFORMANCE INDICATOR (KPI) EVALUATION
SYSTEM**

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1. INTRODUCTION

The measurement and management of performance in higher education institutions is gradually becoming recognized as essential for promoting accountability, quality, and strategic educational development. Traditional performance evaluation methods in universities often rely on subjective assessments and manual reporting. These systems are typically slow, error-prone, and lack transparency, making it difficult for department heads and their superiors to track performance in real time and make informed decisions.

The integration of digital tools, particularly the Key Performance Indicator (KPI) Evaluation System, offers a promising solution. Such systems simplify & modernize the process of setting, tracking, and evaluating administrative performance. By automating evaluation using structured metrics—such as lecturing quality, student supervision, and administrative responsibilities—KPI systems enhance accuracy and consistency.

This literature review examines the current research and practices surrounding KPI automation in academic institutions, with particular attention to the use of web-based systems. It analyzes the technologies, methods, and models used, the effectiveness of these systems in improving institutional performance, and the challenges related to implementation. The review further considers the relevance of these approaches to justify the development of a KPI Evaluation System at the Copperbelt University's School of Information and Communication Technology (CBU-SICT).

2. RELATED WORK AND PREVIOUS SYSTEMS

2.1. Manual vs. Automated KPI Evaluation Systems

Traditional methods of performance evaluation in universities are largely manual relying on paper-based forms, spreadsheets, and personal judgment. These methods are prone to human error, delays, and inconsistencies. Automated evaluation systems, on the other hand, offer enhanced accuracy, consistency, and timeliness. For example, research by Adams and Richardson (2021) highlighted how web-based KPI platforms improved performance visibility and enabled real-time tracking in corporate and academic settings.

Other studies (Zhou et al., 2019) revealed that automation helped reduce administrative burden and improved employee satisfaction through objective assessments. These systems also supported structured feedback between supervisors and employees, aligning performance with the goals of the institution.

Despite advancements in KPI tracking, challenges do persist. Data quality and standardization remain critical concerns, as inconsistent reporting can lead to inaccurate assessments (Fernández-García et al., 2020). Additionally, balancing automation with human judgment is necessary to ensure fairness and contextual understanding in performance evaluations (Jones & Taylor, 2019).

2.2. Real-time Performance Data and Visualization

A significant advantage of KPI Evaluation systems is the ability to integrate real-time data collection and visualization tools. Wang and Patel (2020) emphasized the role of dashboards and analytics in enabling data-driven decision-making. Additionally, the use of automated tracking reduces human error and improves accuracy in reporting (Lee et al., 2018) while the inclusion of visual tools, such as bar charts and performance graphs, have also been shown to contribute to transparency and user engagement.

In educational contexts, institutions have employed performance dashboards to monitor the effectiveness of their teaching, research output, and student outcomes. These dashboards present complex information in comprehensive formats, facilitating prompt managerial interventions when necessary.

2.3. KPI Systems in Educational Institutions

While corporate applications of KPI systems are well documented, literature specifically focusing on higher education is still emerging. Setiawan and Purba (2021) conducted a systematic review of KPI implementation across diverse sectors and identified a gap in university-specific models. Hristov et al. (2022) emphasized the need for education-focused KPIs that reflect academic goals, stakeholder expectations, and institutional missions.

Additionally, Zhao et al. (2024) constructed a KPI-based performance evaluation model for higher education, incorporating the Balanced Scorecard framework to address faculty workload, research, student engagement, and administrative tasks. Their findings support the value of comprehensive, multidimensional evaluation approaches in academic institutions.

2.4. Integrated Educational Management Tool – Adamson University

Doctor (2022) developed and evaluated an Integrated Educational Management Tool at Adamson University. It was designed to assist faculty in conducting exams and managing results. The system was tested using ISO 9126 software quality standard, particularly focusing on usability, reliability, and functionality. It gained a high usability score and was found effective in automating grading-related tasks. The system also supported faculty coordination and data consolidation, enhancing efficiency across academic departments.

2.5. Performance Management System in Tanzanian Higher Education

A study conducted in Tanzania assessed the implementation of a government-mandated performance management system (PMS) across public universities. The system followed the Performance Improvement Model (PIM), which structured employee evaluation through planning, implementation, monitoring, and reviews over periodic intervals. The study highlighted both the structure of the model and institutional efforts to adhere to it, including setting individual targets aligned with national and institutional goals.

2.6. DEMATEL + Linguistic Fuzzy KPI Evaluation Model – Shanghai University

A Chinese research team proposed a hybrid model combining the Decision-Making Trial and Evaluation Laboratory (DEMATEL) method with Linguistic Hesitant Fuzzy Sets (LHFS) to identify and evaluate KPIs in academic institutions. The system was developed and assessed in Shanghai University to account for expert hesitation and the subjective nature of performance measurement. This approach provided a structured framework for linking institutional goals with measurable academic outputs.

2.7. Balanced Scorecard Implementation in Kenyan Higher Education Institutions

Marete (2021) examined the use of the Balanced Scorecard (BSC) framework within a higher education institution in Kenya. The study provided empirical analysis of how BSC principles were used to execute the vision of an institution with measurable academic and administrative performance. The implementation covered multiple fields, including student learning outcomes, faculty productivity, and internal process improvements, providing a foundation for long-term strategic planning.

2.8. System Usability and User Acceptance

For KPI Evaluation systems to succeed, ease of use and user trust are essential. Research by Nielsen (2012) emphasizes that system usability directly influences user satisfaction and adoption rates. In academic settings, where workloads are high and technical learnedness varies, user-friendly designs are critical. Moore (1998) argues that systems designed with user needs in mind increase engagement and reduce errors, reinforcing the importance of user-centric design.

Additionally, as noted by Bwalya and Mutula (2016), ICT system embracement in sub-Saharan Africa must consider infrastructure limitations and provide responsive user support to overcome resistance and technical challenges. Zhao et al. (2023) further suggest that system design should include transparency features to gain stakeholder trust and clarify how KPIs are calculated and applied.

2.9. KPI Management System vs. Excel-Based Evaluation – Malaysian Case Study

A comparative study conducted in Malaysia assessed the effectiveness of a dedicated KPI Management System (KPI-MS) against a traditional Excel-based approach. User satisfaction was evaluated based on system usefulness, information quality, and ease of use. The KPI-MS was found to be more efficient and user-friendly, especially for large datasets and institutional-scale reporting. The system supported centralized performance tracking and was accessible to administrators and lecturers alike.

2.10. Case Study: KPI Implementation at Vasyl' Stus Donetsk National University

Shaulska et al. (2021) conducted a thorough study on the implementation of a KPI based performance management system at Vasyl' Stus Donetsk National University. Over a period of three-years, the university integrated KPIs into its performance evaluation processes to enhance accountability and strategic alignment.

The study exposed key challenges such as resistance to change, complexity in understanding KPI relevance, and communication gaps. Faculty and staff were initially skeptical of the system due to concerns about increased scrutiny and a lack of transparency. Some KPIs were not clearly aligned with individual roles, resulting in confusion and disengagement. Additionally, insufficient communication regarding the purpose and benefits of the system limited stakeholder buy-in.

To overcome these barriers, the university emphasized clear communication, inclusive stakeholder participation in KPI development, and ongoing training. These efforts improved user acceptance and facilitated a more constructive performance culture. The case study highlights the importance of addressing organizational culture alongside technological implementation for successful adoption.

3. SUMMARY

KPI Evaluation Systems in education are essential as they provide a more structured and transparent approach to performance management when compared to traditional methods. By making use of real-time data, structured metrics, and visualization tools, these systems enhance decision-making as well as the overall efficiency of the institution. The literature supports the view that automated evaluation tools reduce subjectivity, increase accountability, and provide valuable insights at both individual and institutional levels.

4. LESSONS LEARNT FROM THE REVIEW

The literature researched underscores the importance of the following:

- Using web-based platforms to improve performance-tracking efficiency and reduce administrative burden.
- Integrating real-time data and dashboards for decision-making.
- Adopting standardized performance criteria to ensure fairness.
- Incorporating analytics to uncover performance trends.

These findings inform the design of the CBU-SICT KPI Evaluation System by supporting the need for role-based access, and visual performance reporting.

5. CRITIQUE OF THE REVIEW

5.1. Web-Based KPI Platforms for Academic and Corporate Environments (Adams & Richardson, 2021)

Strengths:

- Demonstrated the effectiveness of web-based KPI platforms in both academic and corporate environments. Highlighted real-time tracking and visibility, which improves performance transparency.

Limitations:

- Focused mainly on general institutional performance; lacked depth in academic-specific KPI structuring.

5.2. Automation and Administrative Burden Reduction in Performance Evaluations (Zhou et al., 2019)

Strengths:

- Emphasized how automation reduced administrative workload. Encouraged structured feedback between employees and supervisors.

Limitations:

- Concentrated on employee satisfaction but did not address scalability across larger academic departments.

5.3. Data Quality and Standardization in Performance Systems (Fernández-García et al., 2020)

Strengths:

- Identified the need for high-quality, consistent data in automated performance systems.

Limitations:

- Primarily diagnostic in nature; provided fewer actionable solutions to mitigate data inconsistency issues.

5.4. Human Oversight in Automated Evaluation Systems (Jones & Taylor, 2019)

Strengths:

- Addressed the balance between automation and human oversight in evaluation systems.

Limitations:

- Focused on fairness but did not offer a measurable framework for applying human judgment reliably.

5.5. Real-Time Dashboards for Decision-Making (Wang & Patel, 2020)

Strengths:

- Highlighted the use of real-time dashboards and analytics in improving decision-making.

Limitations:

- More relevant to business settings; application to academic performance needed contextual adaptation.

5.6. Reducing Human Error through Automated Tracking (Lee et al., 2018)

Strengths:

- Reinforced how automated tracking reduces human error in reporting.

Limitations:

- Did not explore how the complexity of academic evaluation might affect reporting accuracy.

5.7. Systematic Review of KPI Use Across Sectors (Setiawan & Purba, 2021)

Strengths:

- Provided a comprehensive overview of KPI adoption across various sectors.
- Highlighted the lack of education-specific KPI frameworks especially in universities.

Limitations:

- Did not propose concrete KPI models for academic institutions.
- Lacked case studies or practical implementation examples to support theoretical findings.

5.8. Framework for Academic KPI Development (Hristov et al., 2022)

Strengths:

- Advocated for KPIs that align with academic missions and stakeholder expectations.

Limitations:

- Largely theoretical and did not provide a tested KPI model.

5.9. Transparency and Explainability in KPI Systems (Zhao et al., 2023) & Balanced Scorecard for Academic KPI Evaluation (Zhao et al., 2024)

Strengths:

- Emphasized the need for transparency and explainability in KPI systems. Proposed the Balanced Scorecard model for academia, addressing faculty workload and student engagement.

Limitations:

- Requires significant customization and resource allocation, which may not be feasible in all institutions.

5.10. Integrated Educational Management Tool – Adamson University (Doctor, 2022)

Strengths:

- High usability and reliability (scored 4.76/5 on ISO 9126 usability tests).
- Efficiently automates exams and grading tasks, reducing redundancies.

Limitations:

- Focused mainly on assessment logistics and lacks dedicated KPI evaluation functionality.
- Future integration (e.g., item analysis) was suggested, indicating scope for improvement in KPI depth.

5.11. PMS in Tanzanian Higher Education (2025)

Strengths:

- Conforms to government directives (PIM model), ensuring institutional alignment.
- Includes systematic planning, monitoring, and periodic reviews, covering full performance cycle.

Limitations:

- Staff reported performance goals are not adjusted for academic workload - causing unfair evaluations.
- Low awareness and inconsistent implementation, both of which hindered successful adoption.

5.12. DEMATEL + Linguistic Fuzzy KPI Model – Shanghai University (2020)

Strengths:

- Captures complex KPI relationships and expert uncertainty through DEMATEL + LHFS.
- Provides a quantifiable, structured methodology for KPI identification.

Limitations:

- Requires specialized knowledge to apply, as it is not straightforward for typical academic IT teams.
- Demonstrated at only one university and is unclear if adaptable to other contexts.

5.13. Balanced Scorecard in Kenyan HEI (Marete, 2021)

Strengths:

- Provides empirical evidence linking Balanced Scorecard use with performance gains across institutional fields.

Limitations:

- Broad in scope and as a result, the analysis at the department level may not translate across entire universities.

5.14. Usability and User Adoption Principles (Nielsen, 2012)

Strengths:

- Demonstrated the link between system usability and adoption.

Limitations:

- Did not consider academic environments with varied technical skill levels.

5.15. User-Centric Design Principles (Moore, 1998)

Strengths:

- Supported the value of user-centered design in reducing error and increasing satisfaction.

Limitations:

- Focused more on design principles than specific KPI implementations.

5.16. ICT Adoption Challenges in African Institutions (Bwalya & Mutula, 2016)

Strengths:

- Highlighted challenges unique to ICT adoption in African institutions, including infrastructure limitations.

Limitations:

- Did not address performance management systems specifically, but provided useful regional context.

5.17. KPI-MS vs. Excel System – Malaysian HEI (2012)

Strengths:

- The KPI Management System significantly outperformed the Excel alternative in both usefulness and ease-of-use.

Limitations:

- Based on surveys without performance outcome data, primary measures were subjective user perceptions.

5.18. KPI System at Vasyi' Stus Donetsk National University (Shaulska et al., 2021)

Strengths:

- Offered real-world insight into the implementation of KPI systems in higher education. Addressed cultural and communication issues in adopting new performance frameworks.

Limitations:

- Case-specific findings may not generalize well to other institutions without similar governance structures or resources.

6. CONCLUSION

The reviewed literature reinforces the viability of developing a KPI Evaluation System for CBU-SICT. Their exploration in academic institutions highlights a clear shift toward digital, data-driven approaches for performance management. Studies reviewed consistently show that web-based platforms and real-time tracking tools enhance efficiency, transparency, and informed decision-making. These automated evaluation systems offer clear advantages over manual processes, particularly in tracking and evaluating performance based on objective, measurable indicators. Case studies across Africa, Asia, and Eastern Europe further illustrate how successful implementation depends on contextual relevance, user engagement, and institutional support. Even while challenges related to data quality, privacy, and scalability persist, the studies reviewed consistently show that web-based platforms and real-time tracking tools enhance efficiency, transparency, and informed decision-making.

Overall, the literature suggests that for CBU-SICT, a well-designed web-based KPI platform using C#, React.js, and MySQL—supported by visual dashboards and structured evaluation criteria—can simplify lecturer evaluations, align academic performance with strategic goals, and improve accountability across this specific school in the institution making them essential tools for continuous improvement and institutional excellence.